

Program: Diploma in Microelectronics	
Course Code: 6491B	Course Title: Nanoelectronics
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the basics of nanostructures
- To explain the process used for nano fabrication
- To describe different tools used for the characterization of nanostructures
- To describe various nano devices

Course Prerequisites:

Topic	Course code	Course name	Semester
Fabrication		VLSI Design	4
Ion implantation		Introduction to MEMS	4

Course Outcomes:

After the completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain quantum mechanical effects associated with low dimensional semiconductors.	12	Understanding
CO2	Explain the different processes involved in the fabrication of nanoparticles and nanolayers.	14	Understanding
CO3	Explain the different tools used for characterizing nano layers and particles	16	Understanding
CO4	Explain the working principle of nanoscale electronic devices.	16	Understanding
	Series Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3						
CO3	3						
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain quantum mechanical effects associated with low dimensional semiconductors.		
M1.01	Explain 0D, 1D and 2D nanostructures	3	Understanding
M1.02	Derive the expression for density of states in a 1D nanomaterial.	3	Understanding
M1.03	Compare and contrast triangular, square and parabolic quantum wells.	3	Understanding
M1.04	Solve numerical problems to find whether the given material is a nanometric one.	3	Understanding
Contents: Introduction to nanotechnology, Limitations of conventional microelectronics, characteristic lengths in mesoscopic systems, Quantum mechanical coherence. Low dimensional structures - Quantum wells, wires and dots, Density of states of 1D and 2D nanostructures. Basic properties of square quantum wells of finite depth, parabolic and triangular quantum wells			
CO2	Explain the different processes involved in the fabrication of nanoparticles and nanolayers.		
M2.01	Explain Sol-Gel process for synthesis of nanoparticles.	3	Understanding
M2.02	Explain the different steps involved in CVD process for fabricating nanolayers	4	Understanding
M2.03	Explain Ion Implantation and Diffusion Process	3	Understanding

M2.04	Explain different methods to fabricate nanoparticles	4	Understanding
	Series Test 1	1	

Contents :

Introduction to methods of fabrication of nano-layers: physical vapour deposition-evaporation & Sputtering, Chemical vapour deposition, Molecular Beam Epitaxy, Ion Implantation, Formation of Silicon Dioxide- dry and wet oxidation methods.

Fabrication of nano particle- grinding with iron balls, laser ablation, reduction methods, sol gel, self assembly, precipitation of quantum dots.

CO3	Explain the different tools used for characterizing nano layers and particles		
M3.01	Describe the working principle of STM	3	Understanding
M3.02	Illustrate the working principle of an AFM.	3	Understanding
M3.03	Explain the different emission and interactions between electron beam and the specimen.	6	Understanding
M3.04	Explain the principle of operation of an XRD	4	Understanding

Contents:

Introduction to characterization of nanostructures: Principle of operation of Scanning Tunnelling Microscope, Atomic Force Microscope, Scanning Electron microscope - specimen interaction, Xray Diffraction analysis

CO4	Explain the working principle of nanoscale electronic devices.		
M4.01	Explain the working Principle of MODFETS	3	Understanding
M4.02	Explain Coulomb blockade effect. Illustrate the working of a single electron transistor.	3	Understanding
M4.03	Describe Resonant tunnel effect and the devices	5	Understanding
M4.04	Explain Quantum well laser and quantum dot laser	5	Understanding
	Series Test 2	1	

Contents :

Nanoelectronic devices - MODFETS, Single Electron Transistor, CNT transistors – Properties

of graphene Resonant tunnel effect, RTD, RTT, Hot electron transistors Quantum well laser, quantum dot LED, quantum dot laser

Text/Reference:

T/R	Book Title/Author
T1	J.M. Martinez-Duart, R.J. Martin Palma, F. Agulle Rueda Nanotechnology for Microelectronics and optoelectronics , Elsevier, 2006
T2	W.R. Fahrner, Nanotechnology and Nanoelctronics, Springer, 2005 Reference Books
R1	Chattopadhyay, Banerjee, Introduction to Nanoscience & Technology, PHI 2012
R2	Poole, Introduction to Nanotechnology, John Wiley 2006.
R3	George W. Hanson, Fundamentals of Nanoelectronics, Pearson Education, 2009.
R4	K. Goser, P. Glosekotter, J. Dienstuhl, Nanoelectronics and nanosystems, Springer 2004.